

CSE 464 Project Part 1 - Graph Parser

Name: Andre Damiani de Souza

GitHub: <https://github.com/BrazilDre/CSE-464-2025-adamian6>

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Project Overview

This is a Java program that parses DOT graph files, allows you to add nodes and edges, and outputs graphs as both DOT files and PNG images.

Features

Feature 1: Parse DOT Graphs

- Reads a .dot file and creates a graph
- Methods: `parseGraph(String filepath)`, `toString()`, `outputGraph(String filepath)`

Feature 2: Add Nodes

- Add single or multiple nodes to the graph
- Methods: `addNode(String label)`, `addNodes(String[] labels)`

Feature 3: Add Edges

- Add edges between existing nodes
- Method: `addEdge(String srcLabel, String dstLabel)`

Feature 4: Output Graphs

- Export as DOT file or PNG image
 - Methods: `outputDOTGraph(String path)`, `outputGraphics(String path, String format)`
-

GitHub Commits

Each feature was committed separately:

1. **Initial Setup** - Commit 2b7c361

<https://github.com/BrazilDre/CSE-464-2025-adamian6/commit/2b7c361>

2. **All Features** - Commit c9d4561
<https://github.com/BrazilDre/CSE-464-2025-adamian6/commit/c9d4561>
 3. **Unit Tests** - Commit ffcc21c
<https://github.com/BrazilDre/CSE-464-2025-adamian6/commit/ffcc21c>
 4. **Example Files** - Commit 19d8ddd
<https://github.com/BrazilDre/CSE-464-2025-adamian6/commit/19d8ddd>
 5. **Main Demo Class** - Commit afbe3b7
<https://github.com/BrazilDre/CSE-464-2025-adamian6/commit/afbe3b7>
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Example input:

```
digraph {  
  a -> b;  
  b -> c;  
  c -> d;  
  d -> a;  
}
```

Build the Project

mvn clean package

This compiles the code, runs all 13 tests, and creates a JAR file.

Expected Output:

```

PS C:\Users\damia\OneDrive - Arizona State University\Desktop\CSE 464 - Fall 25\Course Project Part1\graphparser> mvn package
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/Program%20Files/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO] -----< edu.asu.cse464:graphparser >-----
[INFO] Building graphparser 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- resources:3.3.1:resources (default-resources) @ graphparser ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] --- compiler:3.11.0:compile (default-compile) @ graphparser ---
[INFO] Nothing to compile - all classes are up to date
[INFO] --- resources:3.3.1:testResources (default-testResources) @ graphparser ---
[INFO] skip non existing resourceDirectory C:\Users\damia\OneDrive - Arizona State University\Desktop\CSE 464 - Fall 25\Course Project Part1\graphparser\src\test\resources
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ graphparser ---
[INFO] Nothing to compile - all classes are up to date
[INFO] --- surefire:3.1.2:test (default-test) @ graphparser ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnit4PlatformProvider
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running edu.asu.cse464.graphparser.GraphParserTest
[INFO] Tests run: 13, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.392 s -- in edu.asu.cse464.graphparser.GraphParserTest
[INFO] Results:
[INFO] Tests run: 13, Failures: 0, Errors: 0, Skipped: 0
[INFO] --- jar:3.4.1:jar (default-jar) @ graphparser ---
[INFO] Building jar: C:\Users\damia\OneDrive - Arizona State University\Desktop\CSE 464 - Fall 25\Course Project Part1\graphparser\target\graphparser-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.851 s
[INFO] Finished at: 2025-10-27T19:40:00-07:00
[INFO] -----

```

Run the Demo Program

mvn exec:java

This runs the Main class, which demonstrates all 4 features.

Expected Output:

```

PS C:\Users\damia\OneDrive - Arizona State University\Desktop\CSE 464 - Fall 25\Course Project Part1\graphparser> mvn exec:java
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/Program%20Files/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO] -----< edu.asu.cse464:graphparser >-----
[INFO] Building graphparser 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.1.0:java (default-cli) @ graphparser ---
=== CSE 464 Graph Parser Demo ===

Feature 1: Parsing graph from input.dot
Directed graph
Nodes: 4
Edges: 4
a -> b
b -> c
c -> d
d -> a

Feature 2: Adding nodes x and y
Node count after adding: 6

Feature 3: Adding edge x -> y
Edge count after adding: 5

Feature 4: Outputting graph
Created: output.dot and output.png

=== Demo Complete ===
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.564 s
[INFO] Finished at: 2025-10-27T19:23:50-07:00
[INFO] -----

```

